A Computationally Efficient Refinement of the Fundamental Frequency Estimate for the Adaptive Harmonic Model

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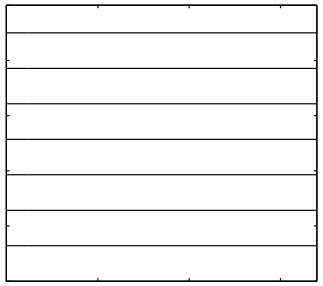
Heraklion, Crete, Greece

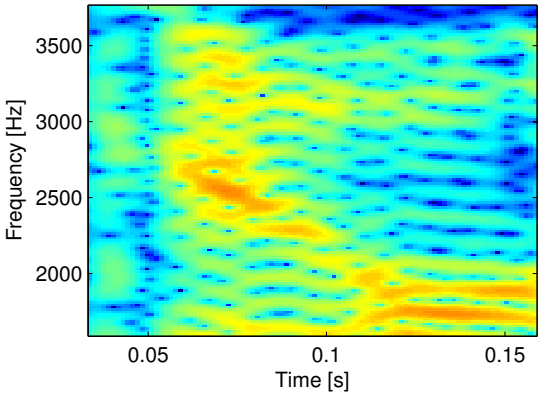
Time-Frequency Representations

DFT $s(t) = \sum_{k=0}^{K} a_k \cdot e^{j\phi_k(t)}$

$\phi_k(t) = k \cdot (2\pi/K) \cdot t$

Constant frequency basis





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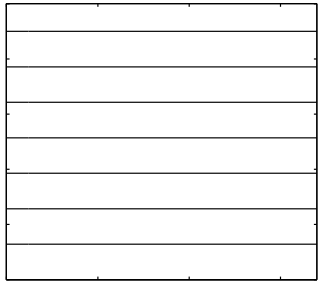

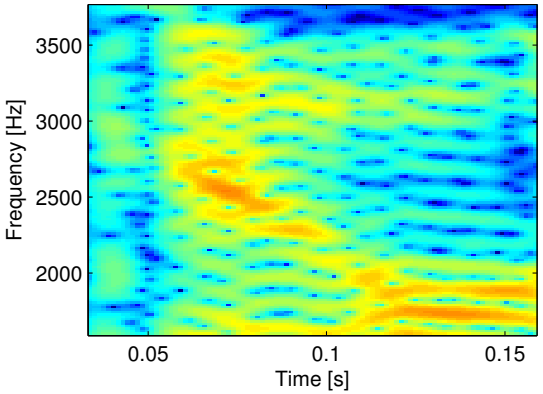

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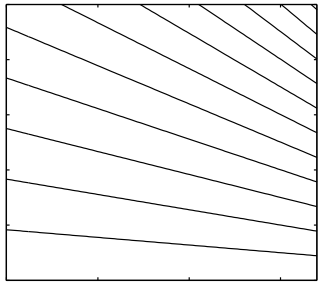
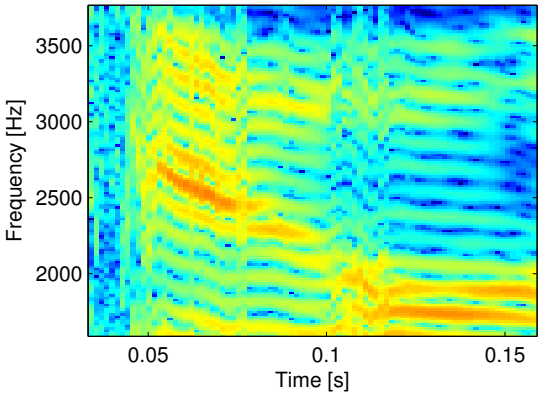
Linear frequency basis





[1] M. Kepesi and L. Weruaga, *Adaptive Chirp-based time-frequency analysis of speech signals*, Speech communication, 2006.

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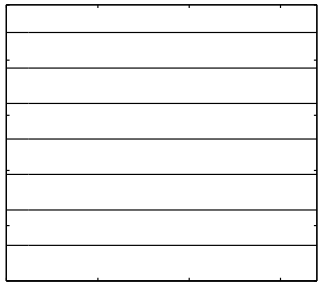

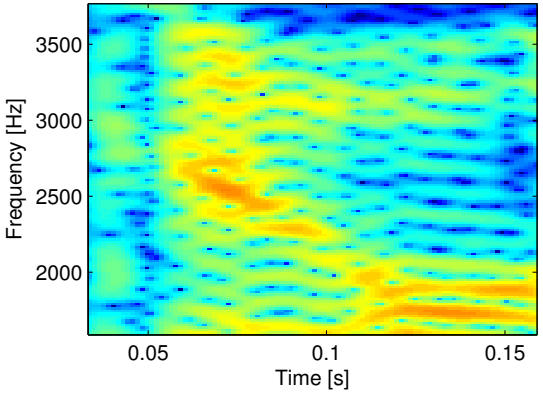

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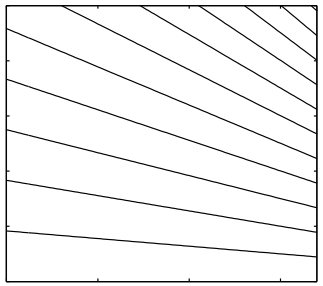

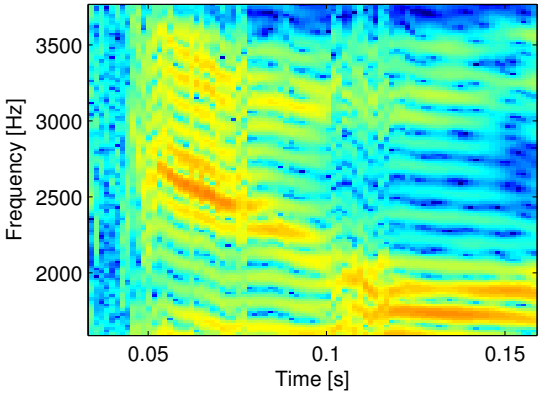

aDFT $s(t) = \sum_{k=0}^{K} a_k \cdot e^{j\phi_k(t)}$

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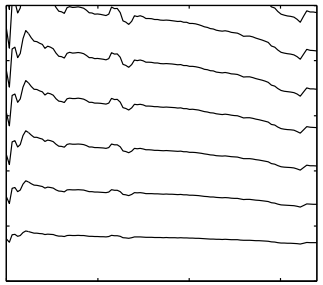
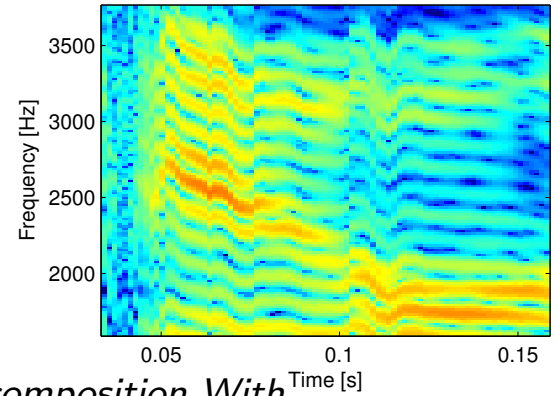
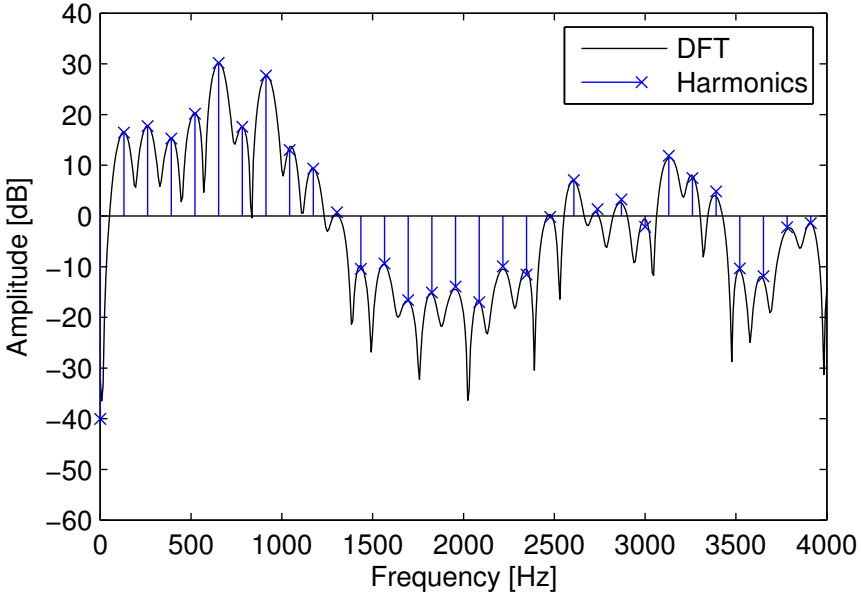
f_0-adapted frequency basis[2]





[2] Y. Pantazis, O. Rosec and Y. Stylianou, *Adaptive AM-FM Signal Decomposition With Application to Speech Analysis*, IEEE Trans. on Audio, Speech, and Language Processing, 2010.

The Sinusoidal and Harmonic Models



Can fit any monophonic signal, but we use it for speech.
The sinusoids can be harmonic, quasi-harmonic, or adaptive ...

The Adaptive Harmonic Model (aHM)

aHM[3] $$s(t) = \sum_{k=-K}^{K} a_k(t) \cdot e^{j(\phi_k(t)+p_k(t))}$$

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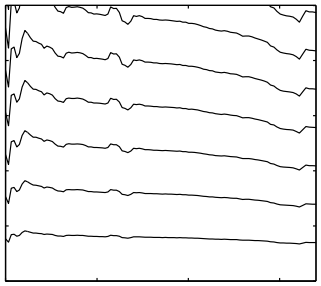

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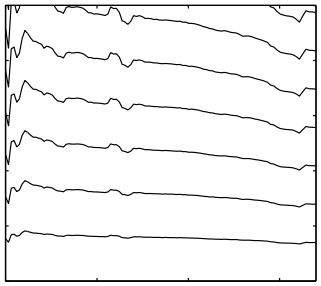

$a_k(t)$ Amplitude: interpolated[2] from $a_k^{t_i}$ at time t_i

$p_k(t)$ Phase: interpolated[3] from $p_k^{t_i}$ at time t_i

$f_0(t)$ Fundamental frequency: interpolated[2] from $f_0^{t_i}$ at time t_i

Parameters at each time t_i: $\{f_0^{t_i}, a_k^{t_i}, p_k^{t_i}\} \quad k \in \{0, ..., K_i\}$

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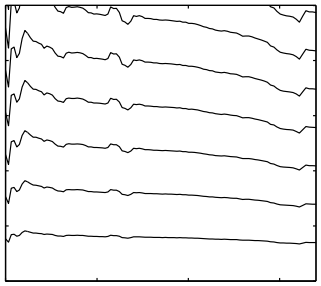

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Full-band representation

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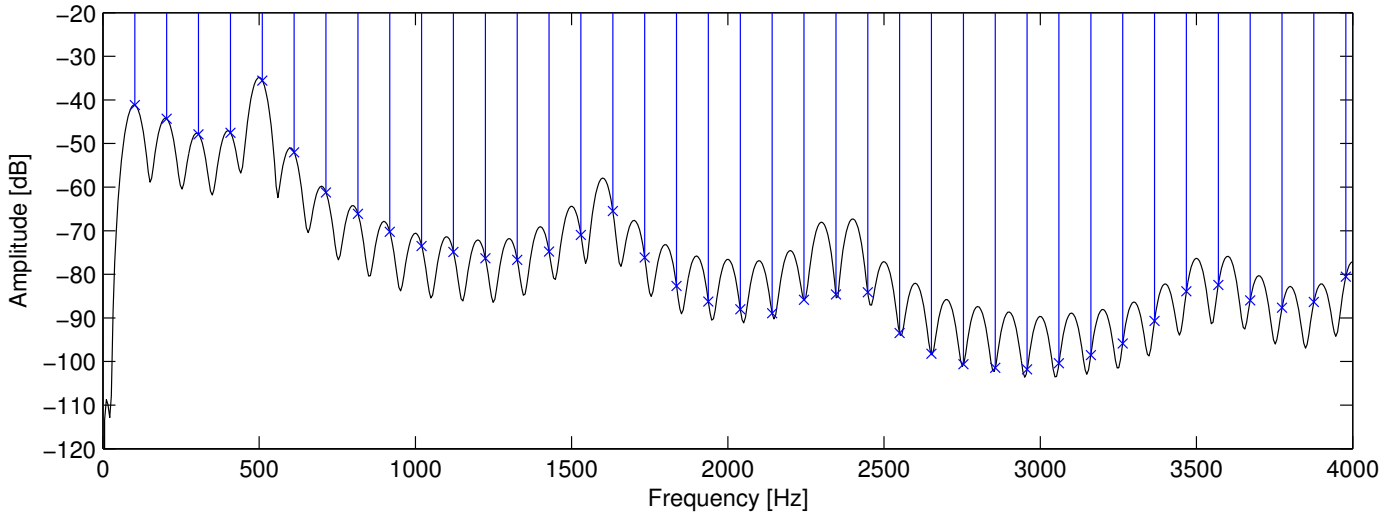
The problem of estimation for full-band representation

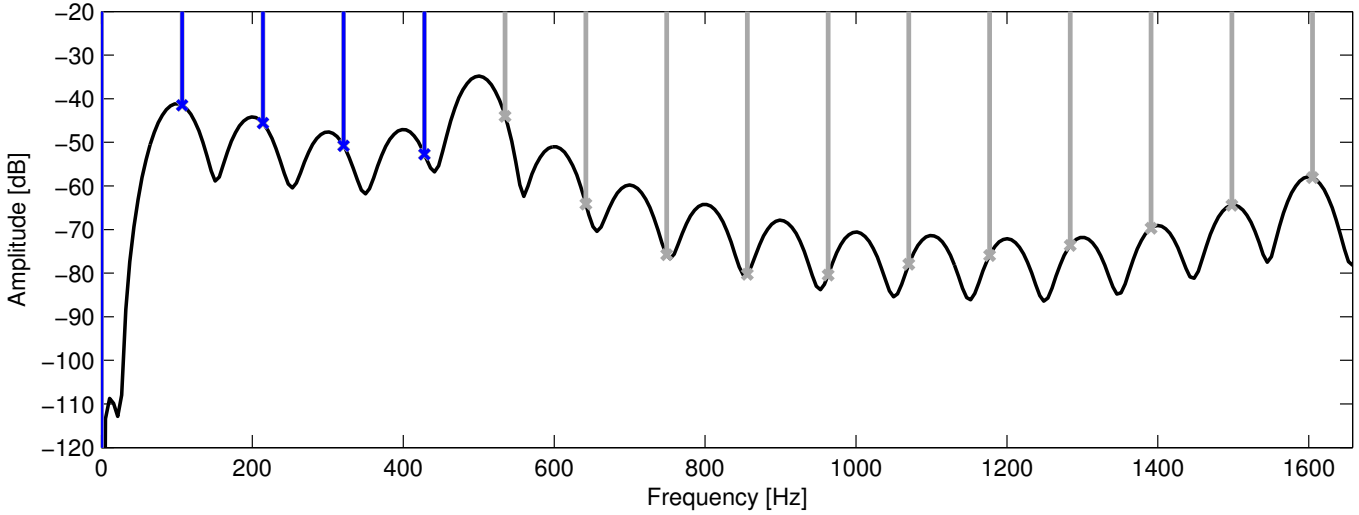
A small f_0 error propagates by multiplication: $f_k = k \cdot f_0$



Problem: How to estimate harmonics up to Nyquist?

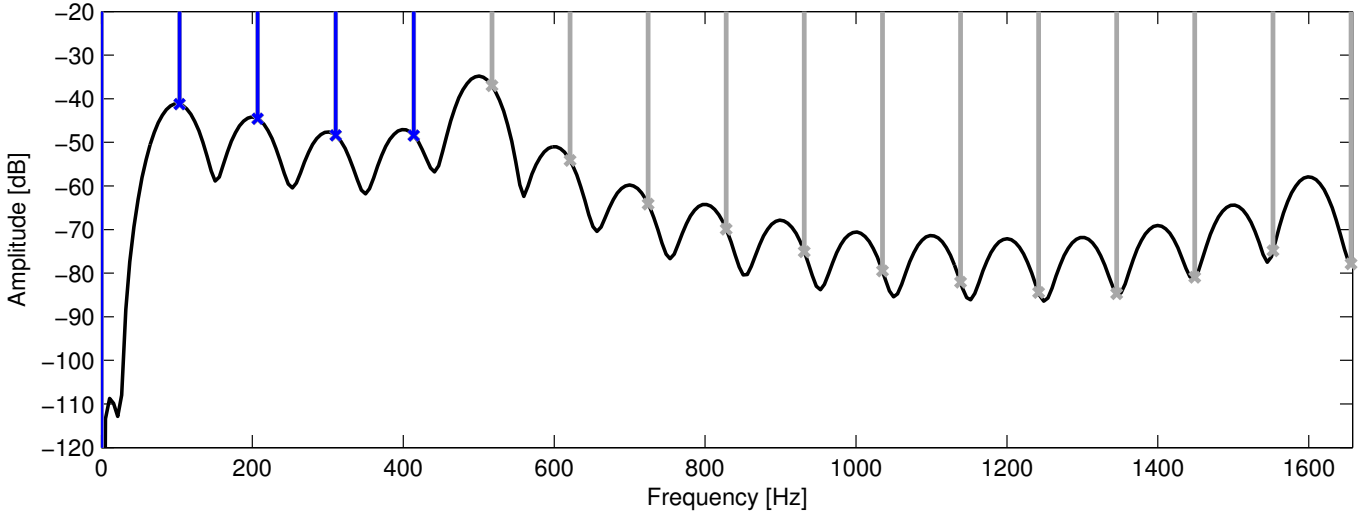
The Adaptive Iterative Refinement (AIR)

First assume the f_0 error is small for low harmonics



The Adaptive Iterative Refinement (AIR)

In previous work, we used QHM[4] to correct f_0



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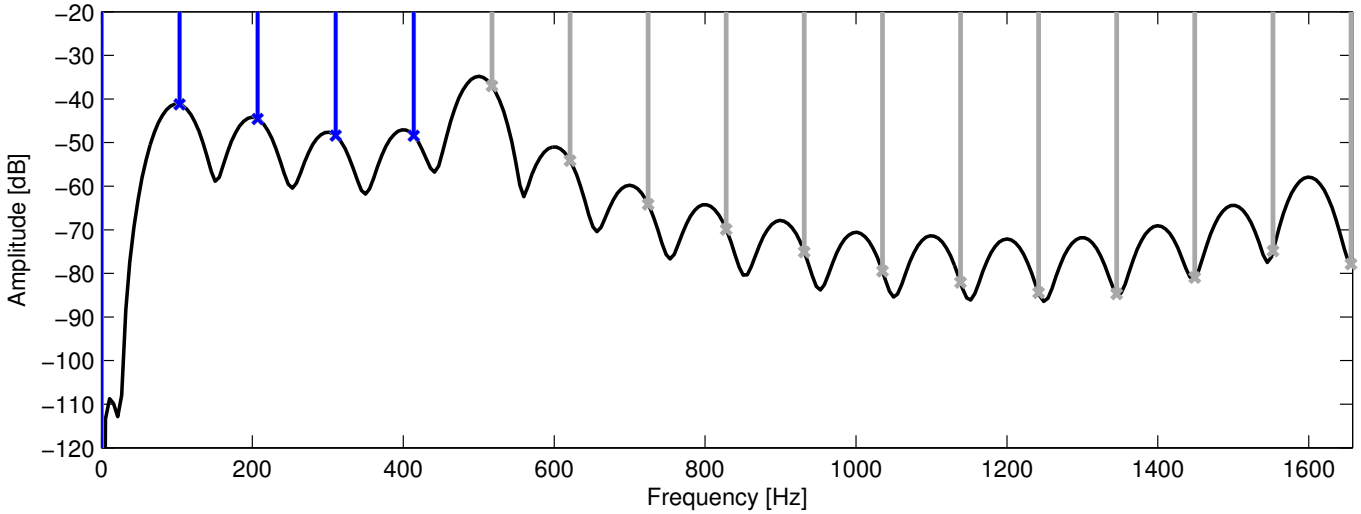


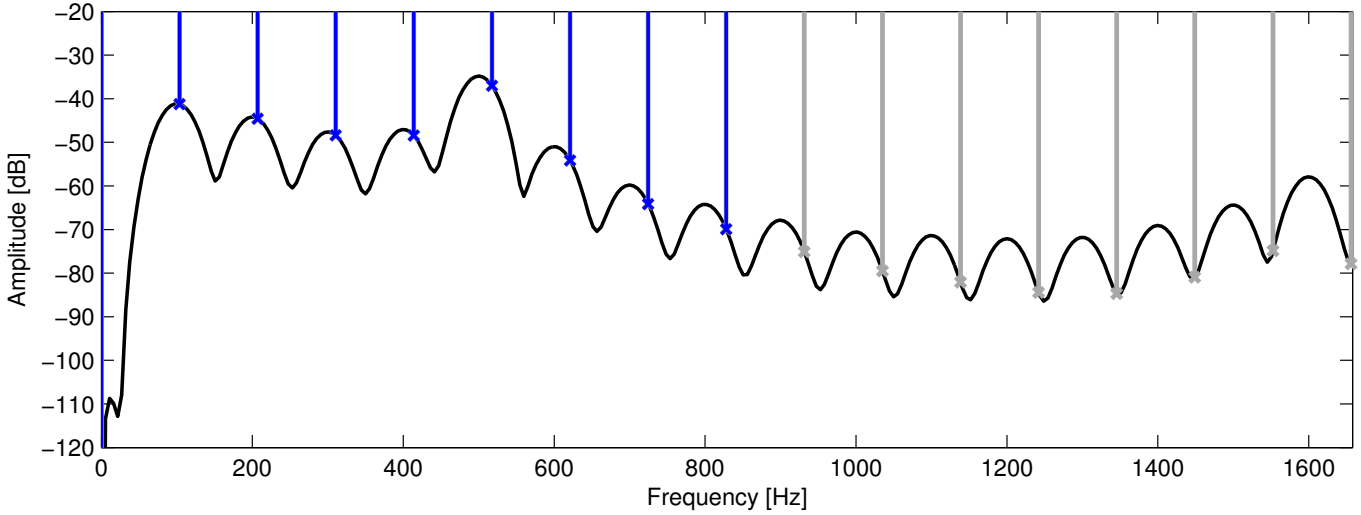
QHM uses a Least Square (LS) solution[4], which is time consuming.
In this paper, we refine f_0 using Peak Picking (PP)[5] from aDFT.

[4] Y. Pantazis, O. Rosec and Y. Stylianou, *Iterative Estimation of Sinusoidal Signal Parameters*, IEEE Signal Processing Letters, 2010.

[5] R. J. McAulay and T. F. Quatieri, *Speech Analysis/Synthesis based on a Sinusoidal Representation*, IEEE Trans. on Acoust., Speech and Signal Processing, 1986.

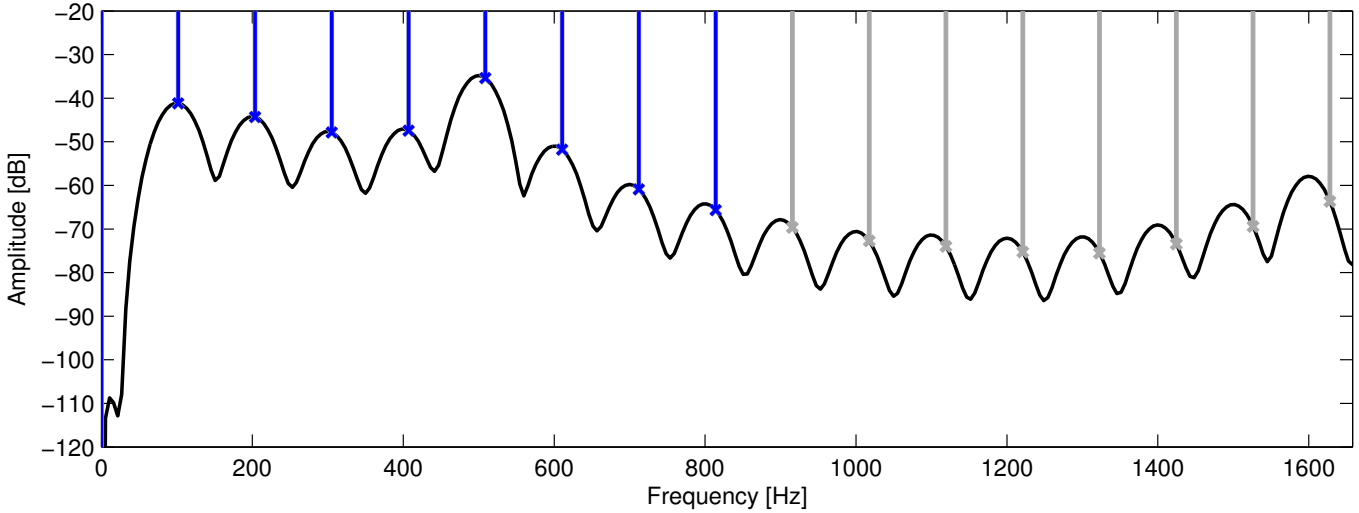
The Adaptive Iterative Refinement (AIR)

After refinement of f_0, we can increase the harmonic level



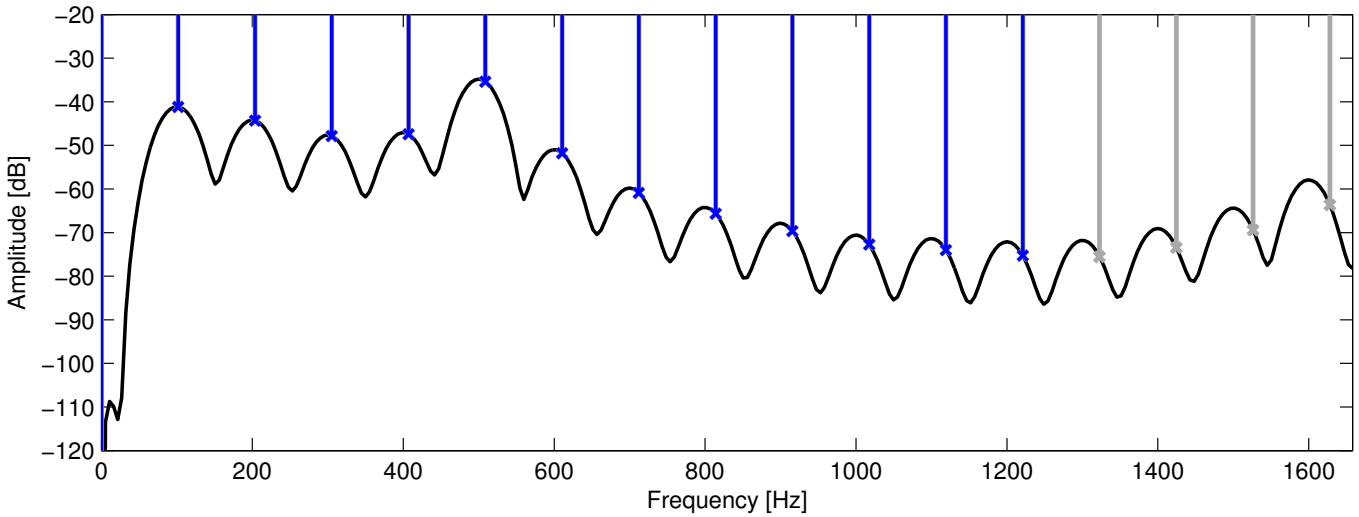
The Adaptive Iterative Refinement (AIR)

Correct f_0 using peak picking from the aDFT



The Adaptive Iterative Refinement (AIR)

Increase the harmonic level



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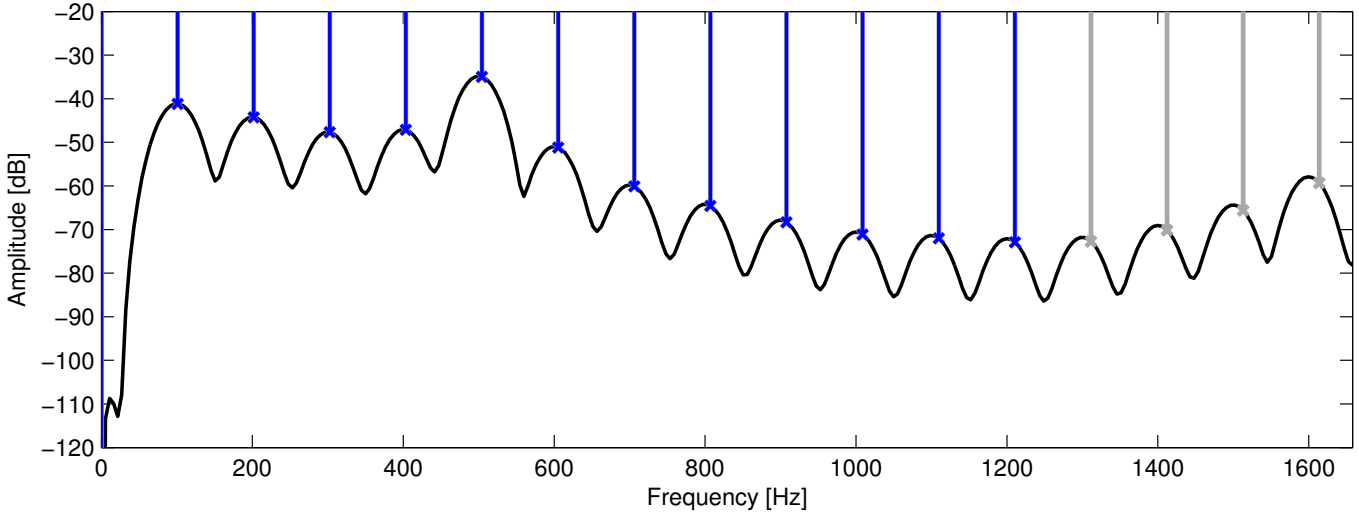
Correct f_0



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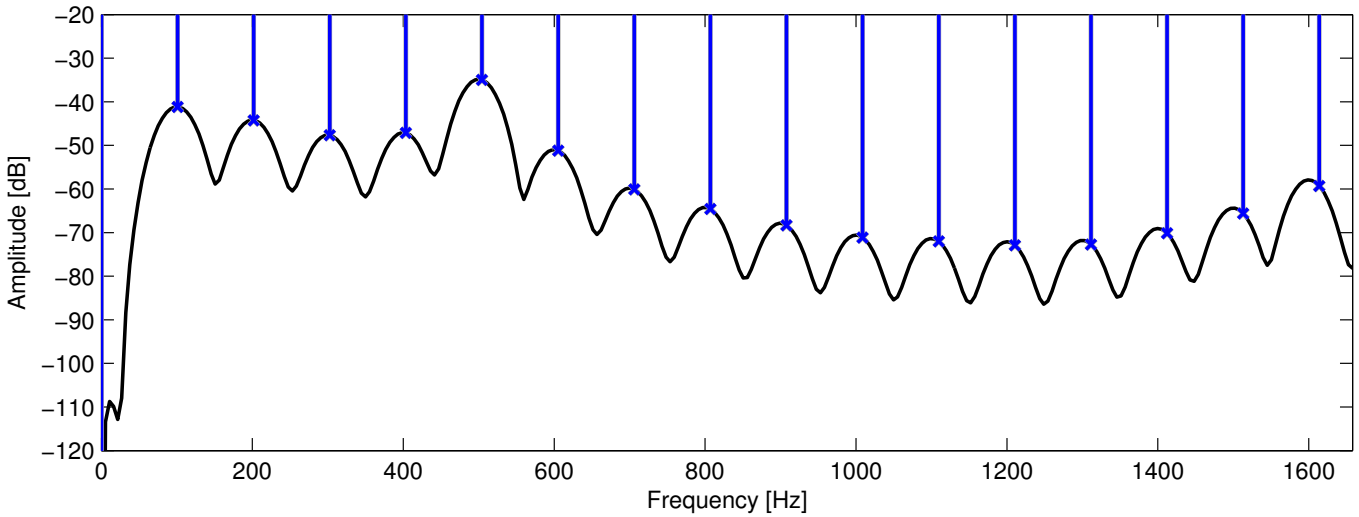

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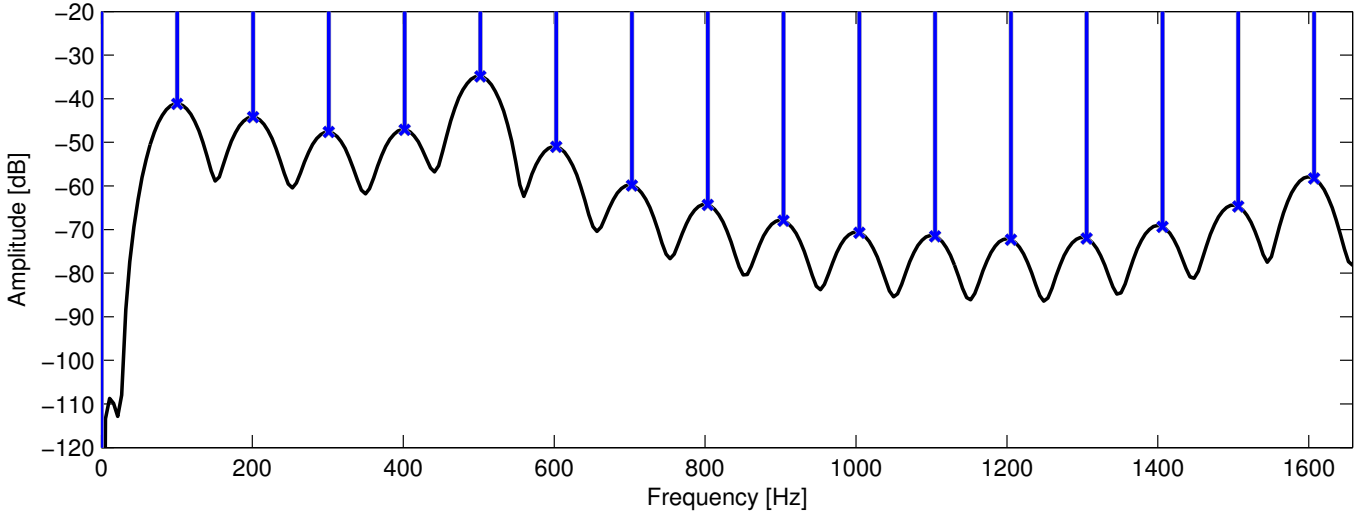
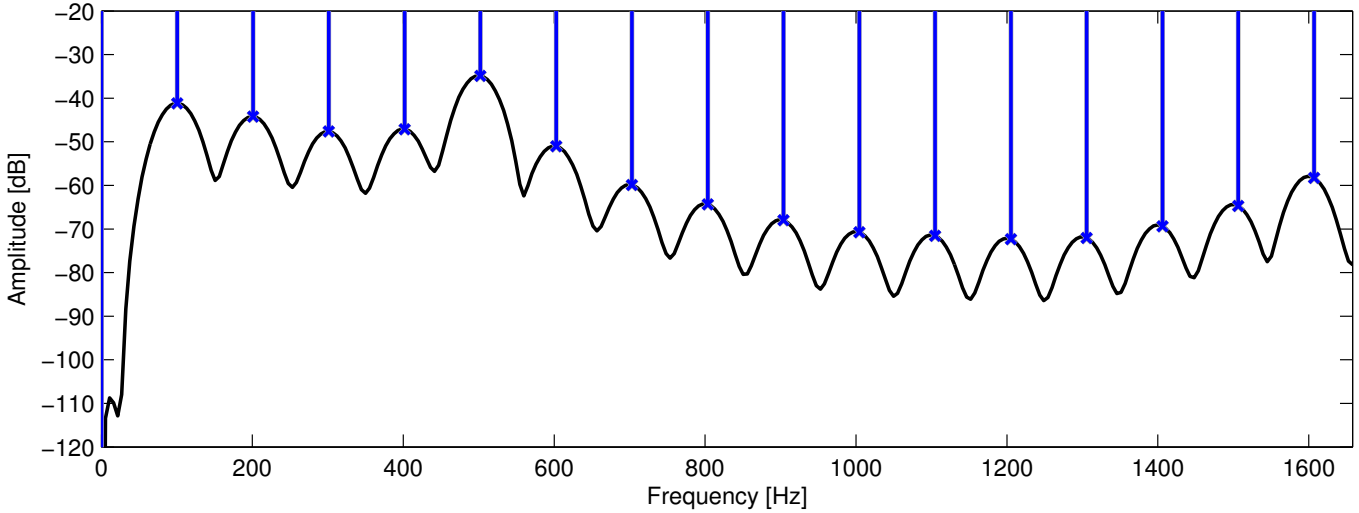


The Adaptive Iterative Refinement (AIR)

Correct f_0 ... until convergence



Evaluation: Computational time

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Male Voices	Female Voices	Average
4.8	6.2	5.5

32 utterances (16 female and 16 male voices in 16 different languages, 2-4s length)

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Average reduction of computational time: 82%

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Perceived quality of the resynthesis by aHM
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`http://www.csd.uoc.gr/~morfi`

Conclusions - Points to remember

In the previous works: Adaptive Harmonic Model aHM

- Frequency tracks adapted to the f_0 curve
- Strict harmonicity
- Full-band, "Full-time", very convenient structural property
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Future works:

Forthcoming paper with estimation of parameters' accuracy and precision.

感謝您的關注

Gracias por su atención

Thank you for your attention

आप अपना ध्यान के लिए धन्यवाद

شكرا لكم على اهتمامكم

Obrigado pela sua atenção

Grazie per la vostra attenzione

Спасибо за ваше внимание

ご清聴ありがとう

మీ శ్రద్ధకు ధన్యవాదాలు

Je vous remercie de votre attention

Σας ευχαριστώ για την προσοχή σας

Dankon pro via atento

please no question about the window